

Task 1 — Document 03: Unique Key Features

Problem Statement ID: PS 26117 / SIH26117 **Title:** Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work **Organization:** Mangalore Refinery and Petrochemicals Limited (MRPL) **Category:** Software | **Theme:** Smart Automation

■ Overview of Innovations & Key Features

The **Sovereign AI Workbench** stands out from traditional cloud AI services and generic chatbot templates through several specialized architectural innovations designed specifically for confidential industrial operations.

Key Feature	Technical Innovation	Operational Benefit
■ 1. Air-Gapped Sovereignty	100% On-Premise Ollama Local Runtime	Zero data exfiltration; complete network air-gap compliance.
■ 2. Integrated Multimodal OCR	Local <code>glm-ocr:q8_0</code> Extraction	Scanned handwritten logs & inspection photos directly searchable.
■ 3. Task Specialist Routing	Model Router (<code>qwen3.5:4b</code> / <code>qwen2.5-coder:7b</code>)	Optimal model dispatch for coding vs. grounded reasoning tasks.
■ 4. Zero-DB Vector Store	Pure Python Cosine Similarity Engine	High-speed dot-product vector search without heavy vector DB setup.
■ 5. Anti-Hallucination Guardrail	Grounded System Prompting & Threshold Rejection	Rejects ungrounded queries ("Information not available in KB").
■ 6. Single-Toggle UI Control	Dynamic Knowledge ON/OFF Switch	Seamless transition between general conversation and strict RAG Q&A.

■ Feature 1: 100% Air-Gapped Data Sovereignty

- **Zero Cloud API Dependency:** Does not rely on OpenAI, Anthropic, Google, or any cloud API endpoints.
- **Air-Gap Certified Execution:** Runs completely offline after initial deployment. Suitable for installation on isolated industrial control system (ICS) networks or air-gapped refinery workstations.
- **Privacy Assurance:** Sensitive operational SOPs, plant blueprints, and maintenance logs never leave the local hardware.

■ Feature 2: Multimodal Image OCR Pipeline (`glm-ocr:q8_0`)

- **Unified Ingestion:** Handles both standard digital documents (`.pdf`, `.txt`, `.md`) and image files (`.png`, `.jpg`, `.jpeg`, `.webp`).
- **High-Precision OCR Model:** Employs `glm-ocr:q8_0`, a specialized quantized vision-language OCR model running via Ollama.
- **Seamless Vector Indexing:** OCR output text is automatically passed to the chunking engine, embedded via `omic-embed-text`, and integrated into the primary Knowledge Base without creating fragmented data silos.

■ Feature 3: Specialized Task Routing Engine

- **Model Specialization over One-Size-Fits-All:** Rather than relying on a single monolithic model, the workbench dynamically dispatches prompts to specialized models:

- **General Reasoning & Grounded RAG:** Powered by `qwen3.5:4b`.
- **Code Synthesis & Scripting:** Automatically routed to `qwen2.5-coder:7b`.
- **OCR Text Extraction:** Dispatched to `glm-ocr:q8_0`.
- **Latency & Resource Optimization:** Using smaller, quantized, high-efficiency models allows high performance on standard workstation hardware.

■ Feature 4: In-Memory Cosine Vector Engine (Zero Heavy DB Overhead)

- **No Heavy Vector Database Needed:** Traditional RAG systems require setting up heavy databases (Milvus, Qdrant, Pinecone), increasing infrastructure complexity.
- **Pure Python Cosine Similarity Engine:** Computes high-speed dot-product and cosine similarity vectors directly in memory, serializing state to a clean JSON index file (`kb_index.json`).
- **Instant Deployment:** Simplifies deployment to a single command (`python main.py / uvicorn main:app`).

■ Feature 5: Grounded RAG & Strict Anti-Hallucination Guardrails

- **Context-Bound Prompt Engineering:** Grounded queries strictly instruct the model to base answers solely on retrieved top-K context chunks.
- **Honest Fallback Guardrail:** When queried about non-indexed topics (e.g., *"What is the refinery emergency shutdown procedure?"* when no shutdown SOP is uploaded), the system explicitly responds:

"That information is not available in the knowledge base."
- **Source Transparency:** Every grounded answer includes source document names and exact cosine similarity matching scores (e.g., `test_sop.txt – Score: 0.9421`).

■■ Feature 6: Responsive Dark-Themed Single-Page Web UI

- **User-Centric Design:** Clean, modern dark-themed web interface built using Vanilla HTML5, CSS3, and JavaScript.
- **Knowledge Toggle (`Knowledge ON/OFF`):** Allows users to seamlessly switch between general conversation mode and grounded document Q&A mode with a single click.
- **Real-Time Drag & Drop File Upload:** Simple interface for uploading PDFs, text files, and inspection images with instant status feedback.